

VIANA DO CASTELO-CHAFE, Portugal

WMO: 085510

Lat: 41.6489N

Lon: 8.8047W

Elev: 160

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.8	36.0	25.8	19.9	43.1	29.1	23.2	42.3	18.4	52.1	15.0	55.2	4.8	100	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	84.6	65.8	80.7	64.4	77.4	64.0	68.9	77.1	67.5	74.5	66.5	72.8	5.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	99.3	70.3	65.6	95.1	69.1	64.5	91.5	68.2	33.1	77.0	32.0	74.7	31.2	73.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.1	10.6	DB	30.5	92.1	2.4	3.3	28.8	94.4	27.4	96.3	26.1	98.2	24.4	100.6	
			WB	28.9	71.5	2.0	2.3	27.5	73.1	26.3	74.5	25.2	75.8	23.7	77.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	48.8	50.0	53.0	56.3	60.1	63.7	66.3	65.3	61.1	54.1	50.5
		DBStd	7.89	4.66	4.25	4.18	4.97	4.97	4.50	4.04	4.21	4.53	5.04	5.03
		HDD50	225	80	48	17	4	0	0	0	0	1	18	57
		HDD65	2819	503	420	371	265	170	75	25	49	140	326	451
		CDD50	3152	42	48	110	193	313	412	504	458	344	142	71
		CDD65	273	0	0	0	4	18	37	65	57	18	0	0
		CDH74	1611	0	0	3	31	159	243	388	420	299	68	0
		CDH80	390	0	0	0	4	37	62	115	104	63	5	0
(14)	Wind	WSAvg	4.6	4.5	5.2	5.4	4.3	4.1	4.1	4.4	4.6	4.2	4.3	5.8
(15)	Precipitation	PrecAvg	56.0	7.5	6.7	4.3	4.4	4.0	2.1	0.9	1.1	3.6	6.1	8.3
(16)		PrecMax	75.0	17.0	19.0	13.6	13.7	10.0	7.4	3.2	3.8	14.4	21.0	23.0
(17)		PrecMin	33.8	0.7	0.2	0.0	0.8	0.2	0.0	0.0	0.0	0.3	0.0	1.1
(18)		PrecStd	10.6	4.2	4.3	3.1	2.9	2.4	1.4	0.9	1.1	3.1	4.1	4.9
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.9	66.6	72.9	77.9	84.8	86.3	89.3	88.0	86.3	80.2	70.1	65.3
(20)		0.4% MCWB	54.0	55.3	55.1	59.5	62.8	66.1	66.5	68.6	66.2	64.4	61.1	54.5
(21)		2% DB	58.5	61.7	67.9	72.8	78.4	81.1	82.9	82.6	80.8	75.4	66.0	61.8
(22)		2% MCWB	55.8	53.2	54.4	59.1	61.4	64.9	66.1	66.0	64.1	62.2	59.9	56.2
(23)		5% DB	57.4	58.9	63.7	69.2	73.4	75.8	77.9	78.2	77.1	71.4	63.5	59.9
(24)		5% MCWB	55.2	53.3	53.3	58.5	61.0	64.0	65.6	65.0	63.5	61.9	59.4	56.4
(25)		10% DB	56.2	57.2	60.6	65.5	69.4	72.3	74.3	75.1	73.8	68.4	61.6	58.2
(26)		10% MCWB	54.1	53.1	52.9	57.2	60.3	63.0	64.8	64.8	63.6	62.2	58.3	55.3
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	58.4	58.1	58.8	62.5	66.2	69.7	70.2	71.9	69.2	67.7	64.4	61.0
(28)		0.4% MCDB	58.8	60.1	64.0	72.0	77.1	81.4	80.9	80.6	78.7	72.2	66.2	61.3
(29)		2% WB	57.1	56.5	57.4	60.9	63.9	67.1	68.1	69.1	67.5	66.3	62.1	58.9
(30)		2% MCDB	57.5	58.7	61.1	69.4	73.1	75.4	76.7	75.4	74.8	69.3	63.8	59.8
(31)		5% WB	55.8	55.2	56.2	59.6	62.5	65.5	66.8	67.5	66.3	65.1	60.7	57.6
(32)		5% MCDB	56.5	57.6	59.8	66.8	70.3	72.6	74.3	73.5	72.2	68.0	62.8	58.8
(33)		10% WB	54.6	53.8	54.9	58.2	61.1	64.1	65.8	66.3	65.0	63.8	59.2	56.1
(34)		10% MCDB	55.7	56.7	58.8	64.1	67.5	70.5	72.4	72.0	70.6	66.9	61.0	57.7
(35)	Mean Daily Temperature Range	MDBR	12.1	13.9	15.0	15.0	15.1	14.6	15.0	15.9	16.2	14.6	12.7	12.7
(36)		5% DB MCDBR	10.8	16.9	21.2	20.9	22.0	20.1	20.8	21.0	21.1	18.9	13.9	12.6
(37)		5% DB MCWBR	8.5	10.5	11.8	10.5	10.2	9.1	8.9	9.1	8.8	9.4	9.0	9.2
(38)		5% WB MCDBR	6.9	13.0	14.5	16.9	17.3	16.3	16.2	16.1	16.2	12.4	10.3	9.4
(39)		5% WB MCWBR	6.6	9.8	9.9	9.5	9.4	8.7	8.4	8.6	8.5	7.8	7.8	8.6
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.346	0.381	0.379	0.386	0.389	0.378	0.390	0.378	0.372	0.344	0.333
(41)		taud	2.480	2.446	2.355	2.372	2.379	2.389	2.411	2.378	2.427	2.455	2.486	2.487
(42)		Ebn at Noon	256	272	274	283	282	281	283	276	272	259	251	247
(43)		Edh at Noon	24	29	35	37	38	37	36	36	32	28	24	22
(44)	All-Sky Solar Radiation	RadAvg	560	886	1239	1654	2007	2186	2213	2000	1560	999	646	506
(45)		RadStd	94	125	186	178	171	146	123	117	108	121	109	77

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+65

Nomenclature: See separate page